FINAL MAIN BUILD · 26 OCTOBER 2026

Space Invaders Presenter Guide

48 scenes. You play the game; you choose when to move on.

Launch with JDK 17 or later

```
powershell -NoProfile -ExecutionPolicy Bypass -File .\scripts\run-game.ps1 -Demo
```

From an IDE, run `engine.Core` with `--demo` and `res` on the classpath.

Presenter controls

CONTROL	ACTION
Back / F6	Previous scene
Restart / F7	Recreate the current scene; purchases remain saved
Next / F8	Continue when ready; no scene timer
F5	Hold or resume prepared cues; gameplay continues
F10	Exit the demo

> Play normally above the footer. Use your configured movement and fire keys. Prepared items usually appear at 8 seconds; boss phase cues run at 12 and 20 seconds. Wait through the normal opening countdown.

> **Build note:** This guide replaces the earlier pending-PR checklist. It follows final main `4007cf6`. Real shop products, rewards, JSON tools, obstacles and boss effects replace the old demo adapters. Starting credits and ranking history are prepared rehearsal data.

PRESENTATION ORDER

Scenes 1–8

Use Next when your explanation is finished. Restart repeats a scene.

#	SCENE	WHAT TO SHOW OR EXPLAIN
1	**Welcome aboard**	Frenchies / VFX / Audio: navigation, stars and menu sounds.
2	**Prepare your mission**	Frenchies: all settings in one menu.
3	**Sound and music**	Chanyoung: adjust BGM/SFX volumes, then press M to toggle mute.
4	**Custom controls**	Frenchies: select an action, press Space, then choose a new key.
5	**Choose your difficulty**	Frenchies / Octopus: Easy, Normal or Hard for campaign gameplay.
6	**Retained settings**	Reopen Controls and Gameplay; confirm cancel/duplicate rejection and retained choices.
7	**Choose your ship**	KimchiBaguette: compare speed, fire rate, lives and projectiles.
8	**Customize your ship**	Frenchies / VFX: use Left/Right to change the ship color.

PRESENTATION ORDER

Scenes 9–16

Use Next when your explanation is finished. Restart repeats a scene.

#	SCENE	WHAT TO SHOW OR EXPLAIN
9	**Two ways to play**	Octopus / Frenchies: campaign progression and endless survival.
10	**Records before the mission**	Chinese can fly: prepared top-ten baseline. Inspect bests and five achievement conditions.
11	**First mission**	Octopus / Friends / VFX / Audio: play using your selected ship.
12	**Campaign HUD and floating points**	Final HUD: stage, lives, score and Credits. Shoot an enemy; watch points and pending Credits update.
13	**Speedster: movement and bursts**	KimchiBaguette: speed 4, two-shot bursts and three lives. Unlock it in the Shop for normal selection.
14	**Heavy Tank: spread shots**	KimchiBaguette: speed 1, three projectiles and five lives.
15	**Damage particles and flashing**	VFX: staged hit at 8s; watch the life loss, red/green flash and player particles.
16	**Third-stage entry: extra life**	Prepared stage-3 entry with 3/4 lives: the final bonus-life rule grants the fourth heart.

PRESENTATION ORDER

Scenes 17–24

Use Next when your explanation is finished. Restart repeats a scene.

#	SCENE	WHAT TO SHOW OR EXPLAIN
17	**Extra life is capped**	Prepared stage-3 entry with Heavy Tank at 5/5 lives: the bonus cannot exceed the cap.
18	**Enemy movement: Classic sweep**	Final game: patterns rotate by stage. Watch smooth movement, aimed shots and acceleration as enemies fall.
19	**Enemy movement: Wave bounce**	Final game: patterns rotate by stage. Watch smooth movement, aimed shots and acceleration as enemies fall.
20	**Enemy movement: Diagonal bounce**	Final game: patterns rotate by stage. Watch smooth movement, aimed shots and acceleration as enemies fall.
21	**Enemy projectiles: 2 lives**	Final adaptive difficulty: newly fired shots use speed 4 (normal).
22	**Enemy projectiles: 1 life**	Final adaptive difficulty: newly fired shots use speed 2 (reduced).
23	**Falling obstacle: movement penalty**	Real red obstacle staged at 8s. Catch it: movement halves for 2s. A shield blocks the slow.
24	**Energy Shield**	Best French: Heavy Tank only: staged shield pickup at 8s, protecting you for 15s.

PRESENTATION ORDER

Scenes 25–32

Use Next when your explanation is finished. Restart repeats a scene.

#	SCENE	WHAT TO SHOW OR EXPLAIN
25	**Twin Cannon**	Best French: staged activation at 8s; shoot to try both cannons.
26	**Hyperfire**	Best French: staged pickup at 8s; hold your fire key to compare fire rates.
27	**Chameleon Hull**	Best French: transformation at 8s; use Up/Down and touch enemies.
28	**Cluster Grenade**	Best French: staged pickup at 8s; press G to launch (if G is unbound).
29	**Boss A: 850 HP, slow**	Final boss mechanics and music; staged phase changes at 12/20s. Defeat it for real score, rewards, VFX and sound.
30	**Boss B: 800 HP, fast**	Final boss mechanics and music; staged phase changes at 12/20s. Defeat it for real score, rewards, VFX and sound.
31	**Orbital Annihilator: stage 7**	Best French: staged pickup at 8s; hold Space to charge the laser.
32	**Endless survival**	Octopus / KimchiBaguette: zigzags and rising difficulty. Next ends this run.

PRESENTATION ORDER

Scenes 33–40

Use Next when your explanation is finished. Restart repeats a scene.

#	SCENE	WHAT TO SHOW OR EXPLAIN
33	**Mission results**	Review your demo score, lives, accuracy and unlocked achievements.
34	**Standard Game Over: no-fire run**	Do not shoot. Prepared lethal hit at 8s; full high-score table prevents name entry. Check Stage 2 Failed and replay.
35	**Stage clear and progression**	Prepared one-enemy stage: defeat it while alive; the real campaign advances to stage 2.
36	**Campaign victory**	Prepared final encounter: one enemy, boss with 50 HP. Clear both alive to show Game Cleared!
37	**Life, shield and slow events**	Heavy Tank: hit at 8s, life at 12s, shield at 16s, obstacle at 20s. Shield blocks the obstacle; watch real events.
38	**Status event bus**	Actual life, shield and slow events from the final game are listed here after gameplay.
39	**Records and achievements**	Chinese can fly: Left/Right switches tabs. This demo uses a separate profile.
40	**High-score medals and icons**	The Diversity Hires: rank medals, hearts, stars, settings and title icons.

PRESENTATION ORDER

Scenes 41–48

Use Next when your explanation is finished. Restart repeats a scene.

#	SCENE	WHAT TO SHOW OR EXPLAIN
41	**Permanent upgrades and ship unlocks**	Final catalog: Hull, Engine and Reloader have three tiers. Speedster costs 200 Credits. Purchases persist.
42	**Credits, pending rewards and save retry**	Real campaign and endless reward services. Optional buttons prepare a one-shot purchase save failure.
43	**Purchased upgrades in campaign**	Start a fresh campaign: bought Hull, Engine and Reloader upgrades affect lives, movement and firing. Endless is unchanged.
44	**Custom levels: JSON import and export**	Octopus: load and validate the bundled example, export a copy to the demo profile, then click Next to play it.
45	**Custom level: two authored waves**	Play the imported enemy grid and both waves. Custom play does not earn official credits, records or campaign unlocks.
46	**Save, reload and restart**	Reload real saved data here. Relaunch with the same demo profile to verify persistence.
47	**Final build and presentation notes**	The demo now uses main's final gameplay. Only scene preparation and navigation are presentation-specific.
48	**End of the demonstration**	Ready for questions. Use Back to revisit a scene, or Finish to close the demo.

PRESENTER REFERENCE

Final rules and remaining limits

AREA	WHAT THE FINAL VERSION DOES
Ship balance	Standard: 4 lives, speed 2, 750 ms shot interval. Speedster: 3 lives, speed 4, 400 ms bursts; unlock for 200 Credits. Heavy Tank: 5 lives, speed 1, 1000 ms spread shots. Only Heavy Tank accepts shields.
Permanent upgrades	Hull adds up to 3 maximum lives, Engine adds up to 3 speed, Reloader reduces the interval by up to 30%. Three tiers cost 100 / 250 / 500 Credits. Effects are captured at the start of a new campaign; Endless stays unchanged.
Rewards and HUD	Campaign rewards use actual enemy classifications and stage completion events. Easy / Normal / Hard clear bonuses are 50 / 100 / 200. Endless grants 1 Credit per defeat. The HUD includes pending earned Credits; purchases use the settled balance.
Bosses and sound	Boss A: 850 HP; Boss B: 800 HP. Production boss selection, phases, defeat points, effects and sounds are active. Boss music plays on boss stages; the urgent last-life track has priority.
Custom levels	The JSON reader, writer, validation and custom gameplay are implemented. Main still has no dedicated import/export menu. The demo provides a panel for a bundled example, export and revalidation. Edit presentation-level.json inside the demo profile and reload to try another definition.
Scope	The final shop contains permanent upgrades and ships. The earlier sample consumable shop has been removed from the demo. Staged Speedster scenes preview its behavior; ordinary Hangar selection still requires the real unlock.

Persistence and rehearsal

Default saves go to `.demo-profile/`, separate from normal player progress. For a clean rehearsal, add `-DemoProfile .\out\final-rehearsal-01` to the launch command. Reuse that path to demonstrate a restart. Back and Restart do not undo purchases.

> **Suggested spoken disclaimer:** “This presentation runs on the final game build. The buttons and prepared encounters help us show each feature. Custom JSON tools are exposed through the demo panel because the main menu has no entry for them.”

This is a source-based integration statement, not a claim about remote PR status. No additional missing PR is assumed. The isolated profile starts with 1000 Credits and prepared high scores.